Semester: Spring

Level: Bachelor
Programme: BE
Course: Microprocessor

Year : 2019
Full Marks: 100
Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

a) What are the major differences between Microprocessor and Microcontroller? Why Harvard architecture is superior to Von-Neumann Architecture? 4+4

b) What are the interrupt pins of 8085 microprocessor? Explain the different methods of handling multiple interrupts 7

a) Write and assembly language program in 8085 to check whether the number stored in memory location D001 H is positive or negative. If positive store it in C001 else store in C002. Comments in program indicate full marks. 8

b) What are the different types of machine cycles in 8085 instructions? Draw the labelled timing diagram for the instruction LDA 30DA H. 2+5

a) Design an Addressing Decoding circuit to interface 2kB RAM, chips each of size 256 bytes with starting address 1001 H. 8

b) Define interrupts. Differentiate between I/O mapped I/O and Memory mapped I/O 7

a) Explain memory interfacing? How do you interface two chips of 4K RAM and one chips of 2K ROM of 8085 with and without fold back addresses, starting address is 8000H? 8

b) Write an 8086 ALP to display the string "POKHARA UNIVERSITY" in reverse order. 7

a) What are the different dedicated interrupt pointers in 8086 microprocessor? Draw and explain the block diagram of 8086 IVT table along with related memory address. 8

b) What is Macro assembler? Differentiate between Macros and Procedure. 7

6. a) What are the significance of assemblers directive in microprocessor? List all the assembler's directive of 8086 microprocessor and explain them with example. 8

b) What is Synchronous and Asynchronous data transmission? Explain the RS232C standard for serial communication. 2+5

7. Write short notes on: (Any two) 2+5

a) SRAM and DRAM
b) RIM and SIM Instruction
c) RS 423A

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